# On the Importance of Initialization and Attention in Sample-Efficient Prompt Tuning

Anonymous ACL submission

## Abstract

Prompt tuning has demonstrated impressive performance across various domains, particularly in few-shot scenarios. However, its underlying mechanisms remain under-explored, and its performance is significantly influenced by initialization methods. In this work, we explore prompt tuning with the FLAN-T5 model. Our results show that dot product similarity between most prompt tokens and label tokens rises during training, and prompt tokens have high attention scores when predicting the label. Based on these insights, we propose the Prompt Tuning Refactoring (PTR) framework. Leveraging our first finding, we initialize prompt tuning tokens directly with label token parameters. To address the second finding, we design a sharp-to-smooth attention mechanism to accelerate model convergence. Additionally, we introduce bridge tokens to help label-initialized tokens effectively aggregate important information. Experimental results in few-shot settings across four datasets demonstrate significant improvements in prompt tuning performance.

## 1 Introduction

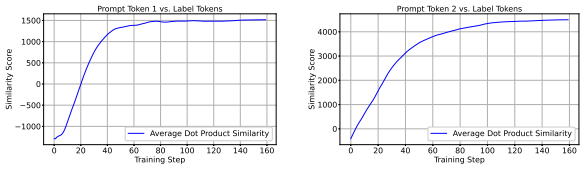
Large language models (LLMs) pre-trained on extensive and diverse general-domain datasets have demonstrated exceptional performance when fine-tuned for specific downstream tasks across various fields, such as natural language processing (NLP) (Devlin et al., 2018; Raffel et al., 2020; Brown et al., 2020; Touvron et al., 2023), computer vision (CV) (Dosovitskiy et al., 2020; Liu et al., 2021, 2022), and multi-model (MM) applications (Radford et al., 2021; Li et al., 2022; Liu et al., 2024). However, as models increase in size, the computational resources required for full fine-tuning become prohibitively expensive. To mitigate this issue, parameter-efficient fine-tuning (PEFT) techniques have gained prominence, updating only a minor fraction of newly introduced parameters while keeping the LLM parameters frozen.

As a key approach in parameter-efficient fine-tuning (PEFT), prompt-tuning has gained prominence with the rise of prompt engineering in large language models. In this study, we conducted an in-depth exploration of prompt tuning, leading to several significant findings:

- Prompt tokens converge towards label tokens, with some showing the highest similarity values among all token representations in the vocabulary after training on a large dataset.
- Prompt tokens show higher attention scores than sentence tokens during decoding.

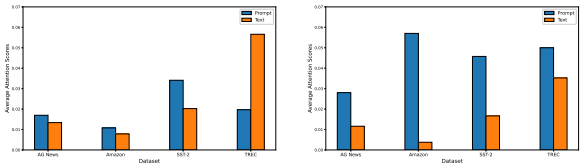
Building on these insights, we propose the Prompt Tuning Refactoring (PTR) algorithm. PTR introduces three key innovations: initializing tunable prompt tokens' parameters directly with diverse label embeddings to expedite convergence, employing a sharp-to-smooth attention mechanism to enhance learning efficiency and gradually reduce dependency on specific tokens, and introducing bridge tokens to help label-initialized tokens effectively aggregate important information. Our contributions can be summarized as follows:

- We discovered that prompt tokens significantly converge to label tokens during training, achieving high attention scores in label prediction.
- We developed the PTR algorithm to enhance information aggregation and passing, accelerate convergence, and improve robustness and generalization.
- Experimental results in few-shot settings across AGNews, SST-2, TREC, and Amazon-5 datasets demonstrate significant performance improvements, substantiating the effectiveness of our proposed methods.




(a) Prompt Token 1 (b) Prompt Token 2

Figure 1: Average Dot Product Similarity Between Prompt Tokens and Label Tokens. This figure tracks the average dot product similarity between each prompt token and label tokens throughout the prompt tuning process.




(a) Pre-Tuning Attention (b) Post-Tuning Attention

Figure 2: Average attention score. This shows the changes in the average cross-attention score before and after prompt tuning, with the importance of the prompt continuously increasing.

## 2 Empirical Study on Prompt Tuning

### 2.1 Prompt-to-Label Convergence

Existing prompt tuning methods (Lester et al., 2021; Chen et al., 2022; Wu et al., 2024) highlight the significance of prompt initialization in determining the performance of language models. Traditional approaches to prompt initialization often neglect to incorporate task-relevant knowledge, leading to a cumbersome and extensive search for an optimal initial prompt within a large space of prompt token embeddings.

Inspired by (Wang et al., 2024), which demonstrates the convergence of prompt tokens' distribution towards image patch tokens in vision prompt tuning, we investigate the relationship between prompt tokens and label tokens during training. We analyze the dot product similarity between prompt token embeddings $\boldsymbol{P} = \{\boldsymbol{p}_1, \boldsymbol{p}_2, \cdots, \boldsymbol{p}_m\}$ and label token embeddings $\boldsymbol{L} = \{\boldsymbol{l}_1, \boldsymbol{l}_2, \cdots, \boldsymbol{l}_n\}$. The average similarity $\boldsymbol{s}_i$ of each prompt token $i$ to all label tokens is calculated as follows:

$$\boldsymbol{s}_i = \frac{\sum_j \boldsymbol{p}_i \cdot \boldsymbol{l}_j}{n} \tag{1}$$

As shown in Figure 1, results from the AG's News Topic Classification dataset (AGNews) (Zhang et al., 2015) reveal a consistent increase in the similarity between most prompt tokens and label tokens throughout the prompt tuning process, indicating that the prompt tokens converge towards the label tokens. Further similarity measurements for other prompt tokens are detailed in appendix A.

### 2.2 Prompt Attention

Recently, attention-based research has received significant interest in in-context learning (Wang et al., 2023; Todd et al., 2024). Labels as anchor (Wang et al., 2023) experimentally validate that in in-context learning, the labels in demonstration aggregate semantic information in shallow layers and control the generation of the final prediction in deep layers.

Then we hypothesize that prompt tokens may play a similar role to label words in in-context learning. We analyze the attention scores in the cross-attention layer to identify key elements in the combined input of prompt and text tokens that are crucial for generating accurate prediction. Representing prompt tokens as $\boldsymbol{p}_i \in \boldsymbol{P}$ and text tokens as $\boldsymbol{t}_i \in \boldsymbol{T}$, with $\boldsymbol{y}$ as the prediction token, We define $S_P$ and $S_T$ as the average cross-attention scores between the prediction token and the respective sets of prompt and text tokens:

$$S_P = \frac{\sum_{\boldsymbol{p}_i \in \boldsymbol{P}} \text{Attn}\left(\boldsymbol{y}, \boldsymbol{p}_i\right)}{L_P} \tag{2}$$

$$S_T = \frac{\sum_{\boldsymbol{t}_i \in \boldsymbol{T}} \text{Attn}\left(\boldsymbol{y}, \boldsymbol{t}_i\right)}{L_T} \tag{3}$$

where $\text{Attn}\left(\cdot\right)$ signifies the cross-attention and $L_P$ and $L_T$ are defined as the lengths of the prompt and text, respectively.

We expand the dataset in Section 2.1 to include Sentiment Treebank Binary (SST-2) (Socher et al., 2013) for sentiment classification, Text REtrieval Conference Question Classification (TREC) (Li and Roth, 2002; Hovy et al., 2001) for question type classification, and Amazon Reviews (Amazon-5) (Zhang et al., 2015) for review sentiment classification. Figure 2 demonstrates a sharp increase in the average cross-attention weight between the prediction token and the prompt tokens after the prompt tuning. This phenomenon likely reflects our preliminary hypothesis that the prompt can aggregate textual information and regulate the generation of predictions throughout the prompt tuning process.

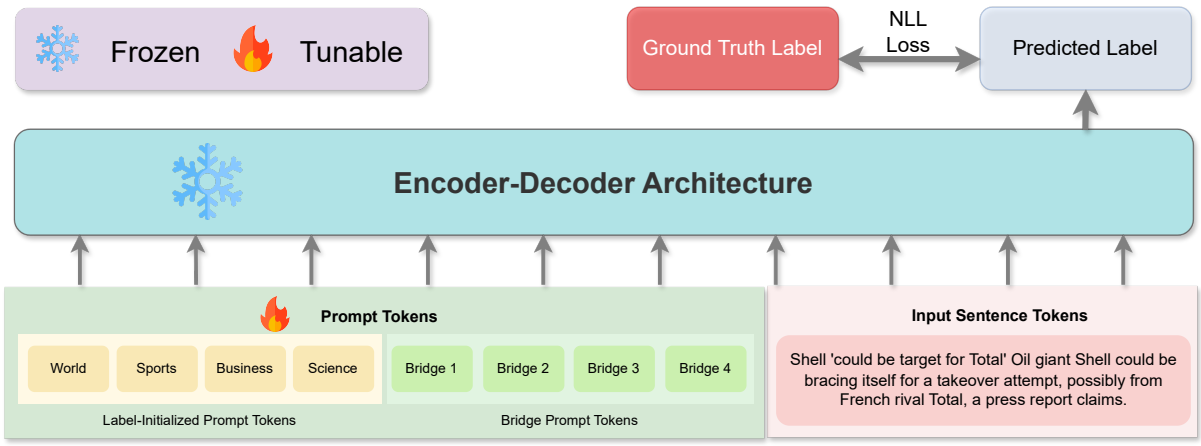



Figure 3: Overview of our architecture. Flame-patterned parameters are tunable and adjustable during the experiment, while snowflake-patterned ones are fixed. At the model's input, labels initialize tunable parameters, which are prepended to the text.

## 3 Prompt Tuning Refactoring

Based on analysis in Section 2.1 and 2.2, we propose an information-driven initialization method and enhance the focus on prompt tokens by incorporating sharp attention with a decaying temperature. Our method is summarized in Figure 3.

### 3.1 Label Prompt Tokens

Drawing on our extensive experimental investigations, we believe that directly initializing prompts with label tokens that are rich in task-specific information can significantly enhance the effectiveness. This method facilitates more effective information exchange between prompts and text, thereby optimizing the influence of prompts on text semantic processing.

### 3.2 Bridge Prompt Tokens

Weak associations between the label token $E_k$ and its keywords $E_t$ can hinder learning efficiency and slow convergence. To address this, we introduce a bridge vector $B_k$, which maintains high similarity with both the label token and its corresponding keywords. As higher similarity enhances information transmission during the attention stage, so replacing low-similarity associations $similarity(E_k, E_t)$ with high-similarity ones $similarity(E_k, B_k)$ and $similarity(B_k, E_t)$ ensures effective information transfer and promotes efficient training and convergence. This process is illustrated in Figure 3, and the detailed implementation of bridge prompt tokens is provided in appendix C.3.

### 3.3 Sharp-to-smooth Cross Attention

Based on the findings in Section 2.2, we propose a method to dynamically adjust the attention to important tokens throughout different training stages. Initially, the model is designed to focus sharply on critical tokens, thereby enhancing the efficiency of parameter learning for these tokens. As training progresses, we aim for a more uniform attention distribution to reduce dependence on specific tokens. This strategy ensures comprehensive attention across all tokens, enabling the model to capture overall patterns and structures in the input data, thus improving its generalization capabilities. Formally, this process is represented as follows:

$$A_i = \frac{\exp(\alpha_i/\tau)}{\sum_j \exp(\alpha_j/\tau)} \tag{4}$$

where $\alpha_i$ represents the original attention score, $A_i$ denotes the adjusted attention score, and $\tau$ gradually transitions from a value less than 1 to a value greater than 1 during training. Further details on the impact of $\tau$ on the distribution are provided in the appendix C.2.

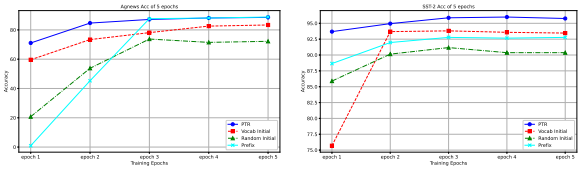
## 4 Experiments

### 4.1 Experimental Setup

We conducted experiments on the four previously mentioned classification datasets, randomly selecting 128 samples from each training set for our analysis and subsequently evaluating performance using the full validation set. We employ flan-t5-large (Chung et al., 2024) as the backbone model.

| Method | #Params | Agnews | SST-2 | TREC | Amazon-5 | Average |
|---|---|---|---|---|---|---|
| Lora | 2.36M | **91.09** | 94.50 | 92.20 | 49.58 | 81.84 |
| Prefix Tuning | 0.98M | 90.82 | 93.12 | 93.20 | **54.78** | 82.98 |
| Prompt Tuning$_{Random\ Init}$ | 0.02M | 89.47 | 95.18 | 79.20 | 27.17 | 72.76 |
| Prompt Tuning$_{Vocab\ Init}$(Default) | 0.02M | 89.72 | **95.41** | 86.60 | 31.63 | 75.84 |
| **Prompt Tuning$_{PTR}$(Ours)** | 0.02M | 90.00 | 95.30 | **95.60** | 51.13 | **83.01** |

Table 1: Classification accuracy results on four datasets over 30 epochs. Prefix-tuning and Lora are two parameter-efficient fine-tuning (PEFT) methods, while random initialization and vocab initialization represent two prompt initialization approaches. We have documented the training parameter requirements for each method.



(a) Agnews (b) SST-2

Figure 4: Convergence rate. This graph records the changes in accuracy on the validation set using different methods over the course of training for 5 epochs.

| Method | Agnews | SST-2 | TREC | Amazon-5 |
|---|---|---|---|---|
| Label | 85.38 | 94.61 | 79.00 | 46.84 |
| Bridge | **87.80** | **94.84** | 86.00 | 46.49 |
| Sharp | 87.11 | 94.27 | **86.20** | **47.04** |

Table 2: Classification results over 5 epochs. "Bridge" refers to adding bridge tokens to label initialization, "Sharp" incorporates sharp attention mechanisms, and "Label" denotes the label-initialized prompt tokens.

More details of the training hyper-parameters can be found in the appendix D.

### 4.2 Main Result

The main results are shown in Table 1. We compared our method with commonly used prompt-tuning initialization methods and other parameter-effective fine-tuning approaches, such as Lora and prefix tuning. We also provide a comparison of the number of trainable parameters across different methods to demonstrate the computational efficiency of our approach. Moreover, we aim to demonstrate the rapid convergence of PTR through the results of experiments conducted over five epochs, and since Lora shows almost no effect within five epochs, we have opted not to include it in the figures.

### 4.3 Ablation Study

To evaluate the effectiveness of Bridge Prompt Tokens and Sharp Attention, we compared our method with label initialization (see Table 2).

### 4.4 Experiment Analysis

As can be found in Table 1, compared to Lora and Prefix Tuning, our method achieves very good results with significantly lower parameter overhead, thereby validating the efficiency of the parameter. The experimental results also indicate that our performance surpasses that of common prompt-tuning initialization methods. We incorporate task-relevant information at the initiation of the prompt, and our use of sharp attention mechanisms enables the model to focus more on the prompt tokens. Furthermore, we validated the performance of various methods over five epochs, shown in Figure 4, and found that PTR converges extremely rapidly. Even in low-resource scenarios, training for just a few steps can yield very good results. As shown in 2, our method improves model performance beyond basic label initialization.

## 5 Conclusion

In this work, we propose PTR, a framework to improve the effect of prompt tuning in low-resource settings. We experimentally validated that during the prompt tuning training process, prompt tokens progressively converge toward label words, and significantly influence the generation of predictions. Therefore, we propose initializing our prompts with a combination of label words and bridge words and employing sharp attention to enable the model to focus more on the prompt tokens from the outset. Extensive experiments have demonstrated that PTR achieves better performance and faster convergence rates in low-resource settings.

## Limitations

In this study, we primarily explored the encoder-decoder (Flan-T5) architecture, discovering that prompt tokens progressively converge toward label tokens during the prompt tuning process and significantly influence the generation of final predictions. However, our investigation did not encompass decoder-only models. We believe that the placement of prompts is also crucial for decoder-only architectures, as it can significantly affect the direction of information aggregation. We hope to conduct more in-depth research on this aspect in the future.

## References


Tom Brown, Benjamin Mann, Nick Ryder, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*.

Xiang Chen, Ningyu Zhang, Xin Xie, Shumin Deng, Yunzhi Yao, Chuanqi Tan, Fei Huang, Luo Si, and Huajun Chen. 2022. Knowprompt: Knowledge-aware prompt-tuning with synergistic optimization for relation extraction. In *Proceedings of the ACM Web conference 2022*.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. 2024. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.

Yuxian Gu, Xu Han, Zhiyuan Liu, and Minlie Huang. 2022. Ppt: Pre-trained prompt tuning for few-shot learning. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.

Karen Hambardzumyan, Hrant Khachatrian, and Jonathan May. 2021. Warp: Word-level adversarial reprogramming. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*.

Eduard Hovy, Laurie Gerber, Ulf Hermjakob, Chin-Yew Lin, and Deepak Ravichandran. 2001. Toward semantics-based answer pinpointing. In *Proceedings of the first international conference on Human language technology research*.

Ting Jiang, Shaohan Huang, Zhongzhi Luan, Deqing Wang, and Fuzhen Zhuang. 2023. Scaling sentence embeddings with large language models. *arXiv preprint arXiv:2307.16645*.

Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*.

Xin Li and Dan Roth. 2002. Learning question classifiers. In *COLING 2002: The 19th International Conference on Computational Linguistics*.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2024. Visual instruction tuning. *Advances in neural information processing systems*.

Ze Liu, Han Hu, Yutong Lin, Zhuliang Yao, Zhenda Xie, Yixuan Wei, Jia Ning, Yue Cao, Zheng Zhang, Li Dong, et al. 2022. Swin transformer v2: Scaling up capacity and resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*.

Colin Raffel, Noam Shazeer, Adam Roberts, et al. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D Manning, Andrew Y Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 conference on empirical methods in natural language processing*.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. 2024. Function vectors in large language models. In *Proceedings of the 2024 International Conference on Learning Representations*.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Jianing Wang, Chengyu Wang, Fuli Luo, Chuanqi Tan, Minghui Qiu, Fei Yang, Qiuhui Shi, Songfang Huang, and Ming Gao. 2022. Towards unified prompt tuning for few-shot text classification. In *Findings of the Association for Computational Linguistics: EMNLP 2022*.

Lean Wang, Lei Li, Damai Dai, Deli Chen, Hao Zhou, Fandong Meng, Jie Zhou, and Xu Sun. 2023. Label words are anchors: An information flow perspective for understanding in-context learning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

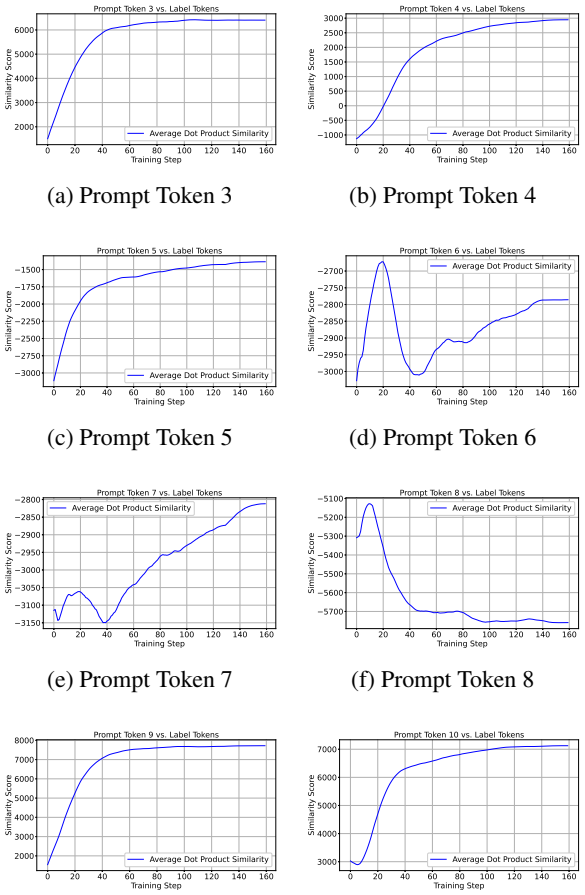
Yuzhu Wang, Lechao Cheng, Chaowei Fang, Dingwen Zhang, Manni Duan, and Meng Wang. 2024. Revisiting the power of prompt for visual tuning. In *Forty-first International Conference on Machine Learning*.

Junda Wu, Tong Yu, Rui Wang, Zhao Song, Ruiyi Zhang, Handong Zhao, Chaochao Lu, Shuai Li, and Ricardo Henao. 2024. Infoprompt: Information-theoretic soft prompt tuning for natural language understanding. *Advances in Neural Information Processing Systems*.

Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. *Advances in neural information processing systems*.

# A Prompt-Label Similarity

Figure 5 illustrates the changes in similarity between other prompt tokens and label words throughout the prompt tuning process.




(a) Prompt Token 3 (b) Prompt Token 4

(c) Prompt Token 5 (d) Prompt Token 6

(e) Prompt Token 7 (f) Prompt Token 8

(g) Prompt Token 9 (h) Prompt Token 10

Figure 5: Average Dot Product Similarity Between Prompt Tokens and Label Tokens. This figure tracks the average dot product similarity between each prompt token and label tokens throughout the prompt tuning process.

# B Related work

## B.1 Soft Prompt Tuning

Pre-trained large language models (LLMs) have demonstrated exceptional performance when fine-tuning downstream NLP tasks (Devlin et al., 2018; Raffel et al., 2020; Brown et al., 2020; Touvron et al., 2023). However, with the model size of LLMs surging, fine-tuning the entire model for each downstream task becomes impractical. As a parameter efficient alternative, soft prompt tuning (Lester et al., 2021; Hambardzumyan et al., 2021) has been proven to be highly effective in NLP downstream tasks, especially in low resource setting (Gu et al., 2022; Wang et al., 2022; Wu et al., 2024), which only updates a minor fraction of newly introduced parameters while keeping parameters of LLMs frozen.

Nonetheless, some recent works (Chen et al., 2022; Wu et al., 2024) have found that the effectiveness of prompt tuning is highly sensitive to prompt initialization. Therefore, it is crucial to identify the appropriate prompt initialization for each downstream task, especially with limited training resources. Integrating downstream task-relevant information into prompt initialization represents a significant challenge in this process.

# C Preliminaries

## C.1 Prompt Tuning

Consider a pre-trained large language model(LLM) with parameters $\theta$ and a downstream task $T$ with training data $\mathcal{D} = \{(\boldsymbol{X}_i, \boldsymbol{y}_i)\}_{i=1}^{N}$, where $\boldsymbol{X}_i \in \mathbb{R}^{s \times e}$ denotes a length $s$ sequence of embedding representations. The objective function of conventional full fine-tuning modifies all the parameters $\theta$ of the model in order to

minimize the negative log-likelihood loss $\mathcal{L}_{NLL}$. The fine-tuning process can be mathematically expressed as follows:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{NLL}(f_\theta(\boldsymbol{X}_i), \boldsymbol{y}_i) \tag{5}$$

In prompt tuning (Lester et al., 2021), a soft prompt $\boldsymbol{P}$, parameterized by $\theta_P \in \mathbb{R}^{m \times e}$, will be prepended to the input text $x$, which will be updated during training while keeping the parameters of the LLM frozen. The objective function in prompt tuning can thus be rewritten as:

$$\min_{\theta_p} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{NLL}(f_{\theta,\theta_p}([\boldsymbol{P}; \boldsymbol{X}_i]), \boldsymbol{y}_i) \tag{6}$$

### C.2 Sharp-to-smooth Attention

Sharp attention is achieved by using smaller temperature coefficients, which amplify the computed attention scores, resulting in more pronounced values after applying softmax. Conversely, smooth attention is achieved by using larger temperature coefficients, leading to more uniform values post-softmax. Figure 6a illustrates the impact of different temperature coefficients on the standard normal distribution, providing an intuitive perception of sharp and smooth distributions. Initially, a smaller temperature coefficient ($\beta$) is used to enhance focus on important tokens. As training progresses, the temperature coefficient is gradually increased, allowing the model to uniformly attend to all tokens. This approach aims to improve the model's generalization and overall performance.

### C.3 Prompt-to-token Similarity

Bridge tokens serve as a bridge between label tokens and their corresponding important tokens with three main functions:

- Aggregate information from label and key tokens, aiding in label prediction control (Figure 6b).
- Transfer information from labels to important tokens (Figure 7a).
- Transfer information from important tokens to labels (Figure 7b).

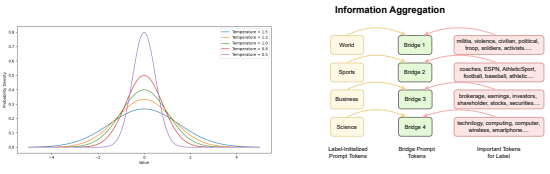
To implement this, we first identify the important tokens related to the label by leveraging the internal knowledge of large models. Inspired by (Jiang et al., 2023), we append "means in one word:" at the end of the input and use the logit of the final token to capture the critical information of the entire sentence. These logits are then categorized into sets based on the labels, and a mean operation is performed within each set. To obtain label-specific important token information, the logits for each label are adjusted by subtracting the average logits of the other labels. Formally, this process is represented as follows:

$$\begin{aligned} L_i &= \text{LLM}(x_i) \\ D_k &= \frac{1}{|S_k|} \sum_{L_i \in S_k} L_i \\ A_k &= D_k - \frac{1}{K-1} \sum_{\substack{j=1 \\ j \neq k}}^{K} D_j \end{aligned} \tag{7}$$

where $L_i$ represents the logit of the output from the large language model (LLM) for input sentence $x_i$ with "means in one word:". $S_k$ is the set containing all logits for label $k$. $D_k$ is the average logit within label $k$, and $A_k$ is the final adjusted logit for label $k$. $K$ denotes the total number of labels.

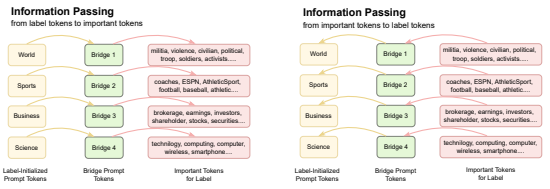
Next, we identify the top $k$ tokens with the highest adjusted logits for each label:

$$\begin{aligned} T_k &= \text{Top-}k(A_k) \\ R_k &= \frac{1}{k} \sum_{t \in T_k} E_t \end{aligned} \tag{8}$$




(a) Different Temperature Coefficients (b) Information Aggregation

Figure 6: Coefficients and Aggregation




(a) Information Passing 1 (b) Information Passing 2

Figure 7: Two Types of Information Passing

$T_k$ represents the set of top $k$ tokens with the highest adjusted logits for label $k$. $R_k$ is the average embedding of these top $k$ tokens, capturing the label-specific important token information.

To ensure the bridge vector has high similarity with both the important tokens and the corresponding label token, we adopt the simplest modeling approach, direct averaging. This approach is both straightforward and effective, allowing us to integrate the information efficiently:

$$B_k = \frac{1}{2}(E_k + R_k) \tag{9}$$

where $E_k$ is the embedding of label $k$, $B_k$ is the final bridge vector representation. This method ensures that the bridge vector effectively combines information from both the label token and the important tokens, facilitating better learning and convergence of the model.

## D Settings

In our experiments, the rank of Lora was set to 4 with a learning rate of 1e-4. For Prefix Tuning, we configured 20 virtual tokens with a learning rate of 1e-2, while Prompt Tuning was implemented with 10 virtual tokens at a learning rate of 0.3.